HeatWise Species Dataset — Field Reference

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Column Groups

A — IDENTITY FIELDS

Column	Type	Source	Description
code	string	seed	Canonical identifier used across DB, ML models, and app (e.g. `tulsi_holy`). Snake-case, unique.
display_name	string	seed	User-facing plant name (e.g. "Holy basil (Tulsi)").
scientific_name	string	seed / GBIF	Binomial or genus-level name (e.g. "Ocimum tenuiflorum").
family	string	GBIF / Perenual	Taxonomic family (e.g. "Lamiaceae").
genus	string	GBIF	Taxonomic genus.
kingdom	string	GBIF	Always "Plantae".
category	string (enum)	derived	`HERB` VEGETABLE SUCCULENT ORNAMENTAL FOLIAGE GRASS CLIMBER SHRUB.
growth_habit	string	derived	Mirrors category for ML CSV column.
cycle	string	Perenual	Annual / Biennial / Perennial / Biannual.
native_range_notes	string	Wikipedia	Short native range description (≤200 chars).
invasive_risk	string	Perenual	HIGH / LOW — whether listed as invasive by Perenual.
hardiness_zone_min	int?	Trefle (not yet)	Minimum USDA hardiness zone. Null until Trefle token provided.
hardiness_zone_max	int?	Trefle (not yet)	Maximum USDA hardiness zone.
max_height_cm	float?	Trefle (not yet)	Maximum expected height in cm.

B — DB BOOLEAN TRAITS (maps to SpeciesCatalog Prisma model)

These drive hard-exclusion filters in the recommendation engine.

Column	Type	Source	Recommendation Engine Role
edible	bool	seed / Perenual	`edibleDominanceTopOpen` — enforces edible-herb ratios in scenarios
flowering	bool	seed / derived	`showyFullSunOrnamental` — demotes showy non-cooling ornamentals in scarce-water setups
pet_safe	bool	seed / Perenual	`HARD_PET_UNSAFE_SPECIES` — hard-excludes if `child_pet_safe_required=1` in scenario
drought_tolerant	bool	seed / derived	`waterScarcityHardExclude` — hard-excludes MED/unknown demand plants unless this is true
heat_tolerant	bool	seed / derived	`shadeSunMismatchHardExclude` — prevents shade-loving plants on full-sun sites
low_maintenance	bool	seed / Perenual	`maintenanceNudge` — boosts low-maintenance plants in low-skill scenarios

C — SUN HOUR FIELDS (maps to SpeciesCatalog)

Column	Type	Source	Description
min_sun_hours	int	seed / Perenual	Minimum direct sun hours tolerated in containers on a roof. Used in `effectiveSunlightPrefNorm()`.

max_sun_hours	int	seed / Perenual			Maximum direct sun hours tolerated.
sunlight_preference	string	Perenual	`FULL`	PART	SHADE — reconciled preference label. Used in `sunMatch()`.

D — ML FEATURE FIELDS (maps to species_features.csv for catalogHybridFallback)

Column	Type	Values								Scoring Function
climate_suitability	string	Pipe-delimited tokens: `HOT_HUMID`	HOT_DRY	HOT_SEMI_ARID	MONSOON_HEAVY	TROPICAL	SUBTROPICAL	MEDITERRANEAN	TEMPERATE	Climate match scoring
water_demand	string	`LOW`	MED	HIGH	`effectiveWaterDemandNorm()`, `waterScarcityHardExclude()`, `scarceWaterPriorityFactor()`					
maintenance_need	string	`LOW`	MED	HIGH	Maintenance scoring nudge					
root_aggressiveness	string	`LOW`	MED	HIGH	Container suitability filter					
pollinator_value	float	0.0–3.0	Ecosystem benefit scoring							
child_pet_safety	string	`SAFE`	CAUTION	UNSAFE	Pet-safe scenario filter (reconciled with pet_safe bool)					
native_support	string	`LOW`	HIGH	Ecological scoring bonus						
container_suitability	string	`POOR`	GOOD	EXCELLENT	Rooftop/balcony eligibility filter					
cooling_contribution	float	0.0–3.0	Primary scoring signal — heat reduction potential							
privacy_contribution	float	0.0–3.0	Privacy benefit scoring							
drought_tolerance	string	`LOW`	MED	HIGH	Legacy vocab field — mirrors drought_tolerant bool					

E — COOLING CONTRIBUTION SCORING GUIDE

Score	Meaning	Typical Plants
3.0	Maximum cooling — dense wall coverage	Climbers (Bougainvillea, Money plant, Thunbergia)
2.8	Excellent — high ET + large canopy	Tropical foliage with high watering
2.5	Very good — grass density or shrub coverage	Lemongrass, Bamboo, dense Shrubs
2.0	Good — moderate foliage + partial shade	Hibiscus, Ornamentals with full sun
1.5	Moderate — herbs, small edibles	Vegetables, most herbs
0.9	Low — sparse leaf area	Succulents, Cacti
0.5	Minimal — decorative only	Flowering bulbs

F — IMAGE FIELDS

Column	Description		
image_url	Full-resolution image URL (Perenual preferred, Wikimedia fallback)		
image_thumbnail	Thumbnail URL for app display		
image_license	License string (e.g. "CC BY-SA 4.0")		
image_source	Origin: `perenual`	wikipedia	wikimedia
image_local_path	Relative path to downloaded image in species_images/		

G — PROVENANCE FIELDS

Column	Description
source_perenual	yes/no — data fetched from Perenual API
source_gbif	yes/no — taxonomy confirmed via GBIF
source_wikipedia	yes/no — description pulled from Wikipedia
source_seed	yes/no — present in curated HeatWise seed (always "yes")
confidence	0.0–1.0 — how many sources corroborated this record

Selection Parameters — How Species Are Chosen for Recommendations

The recommendation engine applies these parameters as filters in order:

- **Hard Exclusion — Sun Mismatch** (`shadeSunMismatchHardExclude`)
- Exclude if `sunlight_preference = FULL` and site sun_hours < 2
- Exclude if `min_sun_hours > site_sun_hours + 0.75`
- **Hard Exclusion — Water Scarcity** (`waterScarcityHardExclude`)
- Exclude `water_demand = HIGH` unconditionally when `water_availability = scarce`
- Exclude `water_demand = MED` unless `drought_tolerant = true`
- Exclude unknown demand unless `drought_tolerant = true`
- **Hard Exclusion — Pet Safety** (`HARD_PET_UNSAFE_SPECIES`)
- Exclude all species where `pet_safe = false` when scenario has `child_pet_safe_required = 1`
- **Container Suitability Filter**
- Exclude `container_suitability = POOR` on rooftop/balcony space types
- **Priority Boosts / Demotions** (`scarceWaterPriorityFactor`)
- `water_demand = LOW` + `drought_tolerant = true` → **1.30×** boost
- `water_demand = LOW` alone → **1.12×** boost
- `water_demand = MED` + `drought_tolerant = true` → **0.90×** demotion
- Showy ornamental (non-edible + flowering + full-sun + `pollinator_value ≥ 3`) → **0.58×** demotion in scarce-water
- **Scoring** — weighted sum:
- `cooling_contribution` × cooling weight
- `sunMatch(sunlight_preference, site_bucket)` × sun weight
- `pollinator_value` × ecosystem weight
- `container_suitability` bonus
- `ml_weight` multiplier (per-species model output)

Dataset generated by HeatWise species scraper — scripts/scrape-species-dataset.mjs